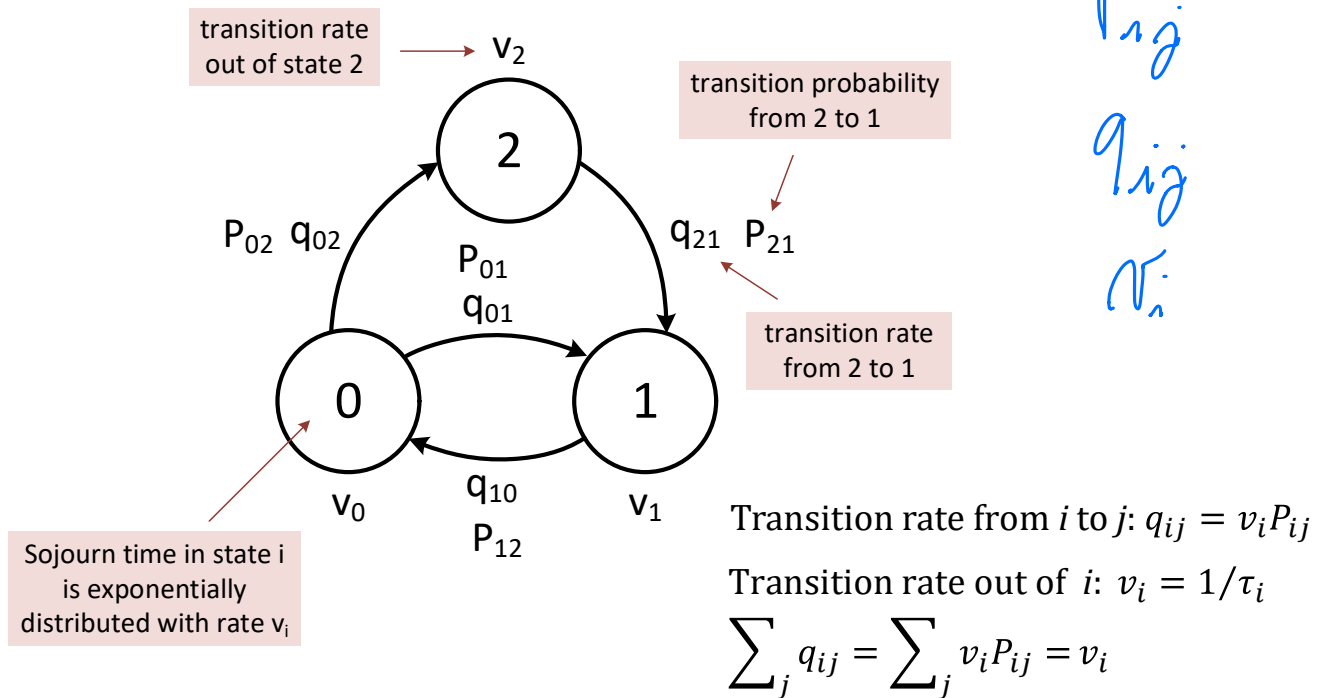


Continuous-time Markov chains

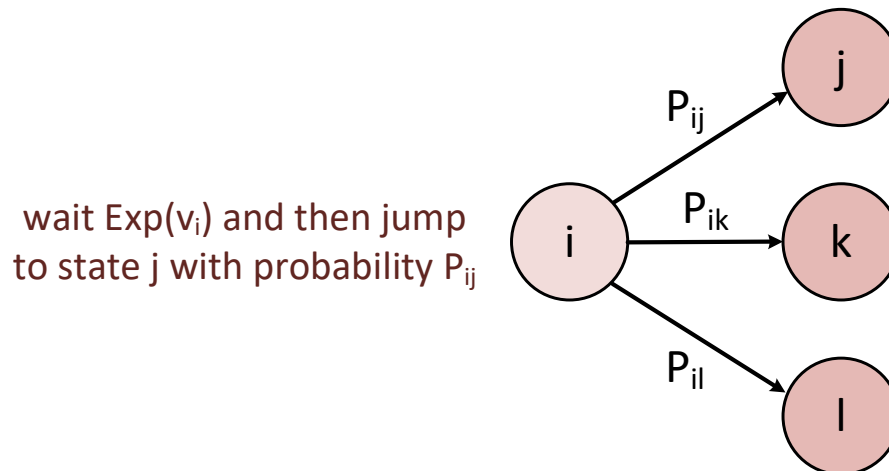
Continuous-time Markov chains

- Continuous-time Markov chains (CTMC) are the continuous-time analogue of DTMCs.



Continuous-time Markov chains

- VIEW 1 - A CTMC is a stochastic process where every time the process enters state i
 - The amount of time the process spends in state i before making a transition is exponentially distributed with some rate v_i .
 - When the process leaves state i , it enters state j with some probability P_{ij} independent of the time spent in state i (NOTE: $P_{ii} = 0, \sum_i P_{ij} = 1$).



Continuous-time Markov chains

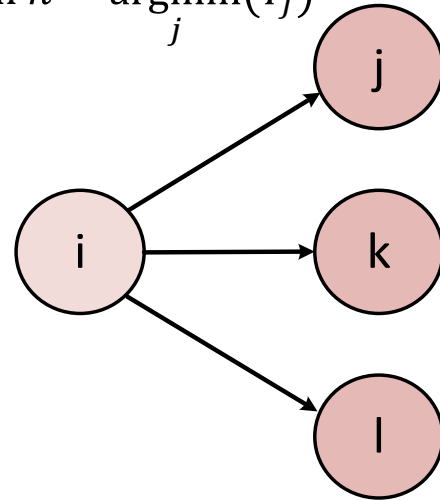
- VIEW 2 - Let T_j represent the time to transition from state i to state j , and τ_i be the minimum of these times, i.e. $\tau_i = \min_j (T_j)$. In a CTMC, T_j is exponentially distributed with rate $q_{ij} = v_i P_{ij}$, τ_i is the time until the Markov chain leaves state i , and the next state n is the state for which T_j was the minimum $n = \operatorname{argmin}_j (T_j)$

wait until first event and
jump to state of that event

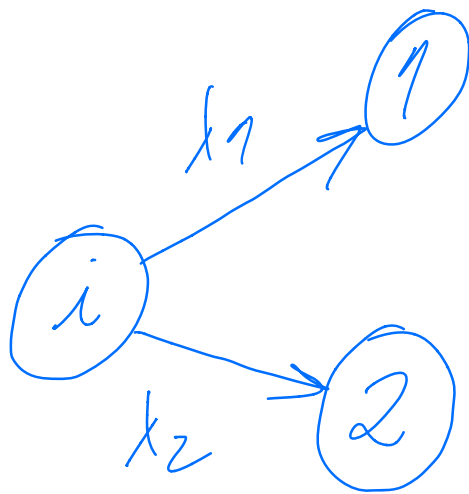
$$T_j \sim \operatorname{Exp}(P_{ij}v_i)$$

$$T_k \sim \operatorname{Exp}(P_{ik}v_i)$$

$$T_l \sim \operatorname{Exp}(P_{il}v_i)$$



$\tau_i = \min_j (T_j)$ is exponentially distributed?



$$\lambda_1 = p_{i1} \nu_i$$

$$\lambda_2 = p_{i2} \nu_i$$

$$T_1 \sim \text{Exp}(\lambda_1)$$

$$T_2 \sim \text{Exp}(\lambda_2)$$

$$\tau_i = \min(T_1, T_2)$$

$$P(\tau_i \leq \tau) = 1 - P(\tau_i > \tau)$$

$$= 1 - P(\min(T_1, T_2) > \tau)$$

$$\begin{array}{c} | \quad | \quad | \rightarrow \\ \tau \quad T_1 \quad T_2 \end{array} = 1 - P(T_1 > \tau, T_2 > \tau)$$

$$= 1 - P(T_1 > \tau) P(T_2 > \tau)$$

$$P(T_1 > \tau) = e^{-\lambda_1 \tau} \quad P(T_2 > \tau) = e^{-\lambda_2 \tau}$$

$$= 1 - e^{-\lambda_1 \tau} \cdot e^{-\lambda_2 \tau}$$

$$= 1 - e^{-(\lambda_1 + \lambda_2)\tau}$$

$$\tau_i \sim \text{Exp}(\lambda_1 + \lambda_2)$$

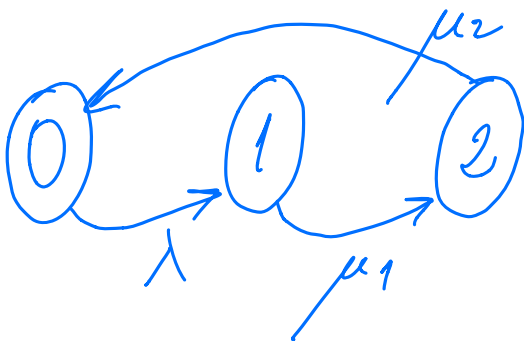
$$T_i = \min(T_1, T_2, \dots, T_n)$$

$$T_j \sim \text{Exp}(\lambda_j), \quad j=1, 2, \dots, n$$

$$T_i \sim \text{Exp}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$$

Continuous-time Markov chains - example

- Consider a shoeshine shop consisting of two chairs, chair 1 and chair 2. A customer upon arrival goes initially to chair 1 where his shoes are cleaned. After this is done the customer moves on to chair 2 where the shoes are polished. The service times at the two chairs are assumed to be independent and exponentially distributed with rates μ_1 and μ_2 , respectively. Suppose that potential customers arrive in accordance with a Poisson process having rate λ and that a potential customer will only enter the system if both chairs are empty. This system can be analyzed as a continuous-time Markov chain?



$$P_{01} = 1$$

$$P_{12} = 1$$

$$P_{20} = 1$$

$$q_{01} = \lambda = \nu_0$$

$$q_{12} = \mu_1 = \nu_1$$

$$q_{20} = \mu_2 = \nu_2$$

Limiting state probabilities

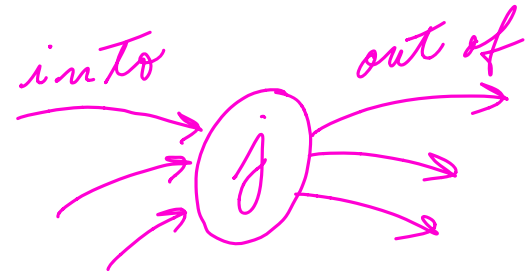
- Limiting state probabilities exist if the CTMC is irreducible and positive recurrent.
- Can be calculated through the **balance equations**:

$$\pi_j v_j = \sum_i \pi_i q_{ij}, \quad \sum_i \pi_i = 1$$

where v_i is the (total) rate at which the process leaves state i , and $q_{ij} = v_i P_{ij}$ is the rate at which the process makes a transition from state i to state j - instantaneous transition rate.

- The probability π_i can be interpreted as the proportion of time that the process spends in state *i*

Balance equations

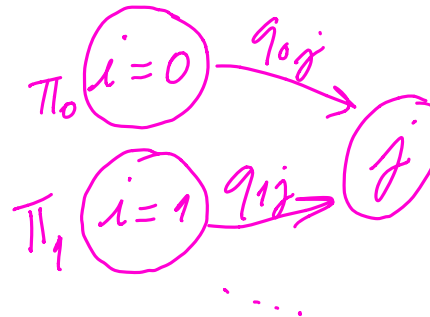


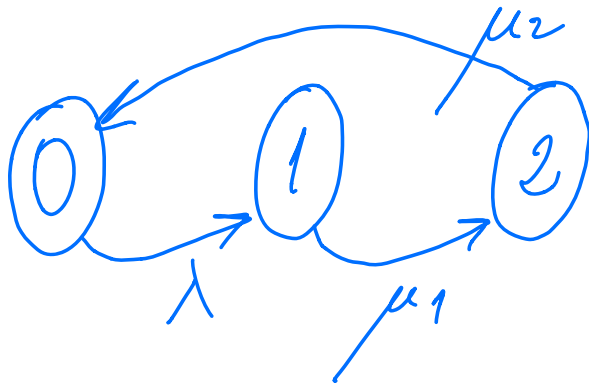
- In any interval $(0, t)$ the number of transitions into j must equal, up to a difference of one, the number of transitions out j . Thus, in a relatively long period,

rate at which the process **leaves** state j
 = rate at which the process **enters** state j

- Moreover,

$\pi_j v_j$ = rate at which the process **leaves** state j
 $\sum_i \pi_i q_{ij}$ = rate at which the process **enters** state j





$$\pi_j \nu_j = \sum_i \pi_i q_{ij} \\ j=0,1,\dots$$

$$0: \quad \pi_0 \lambda = \pi_2 \mu_2 \quad *$$

$$1: \quad \pi_1 \mu_1 = \pi_0 \lambda \quad *$$

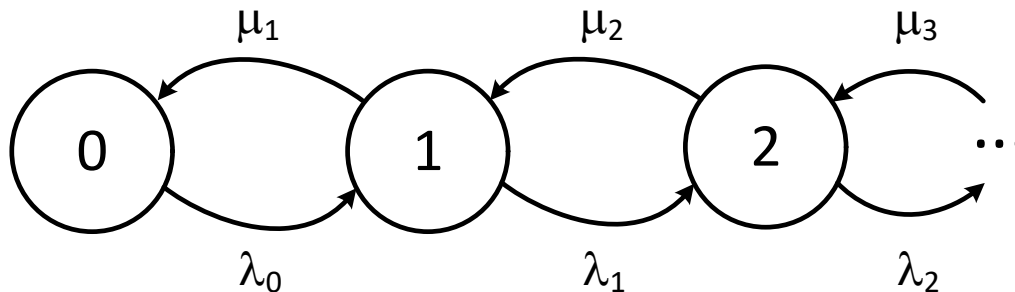
$$2: \quad \pi_2 \mu_2 = \pi_1 \mu_1$$

$$\pi_0 + \pi_1 + \pi_2 = 1 \quad *$$

Birth-death processes

- Consider a system for which the state at any time corresponds to the number of persons in the system at that time. Suppose that if the system has n persons, new persons arrive at an exponential rate λ_n (i.e. the time until the next arrival is exponentially distributed with mean $1/\lambda_n$) and persons leave the system at an exponential rate μ_n (i.e. the time until the next departure is exponentially distributed with mean $1/\mu_n$).
- This is a **birth-death process**. Is this a CTMC?

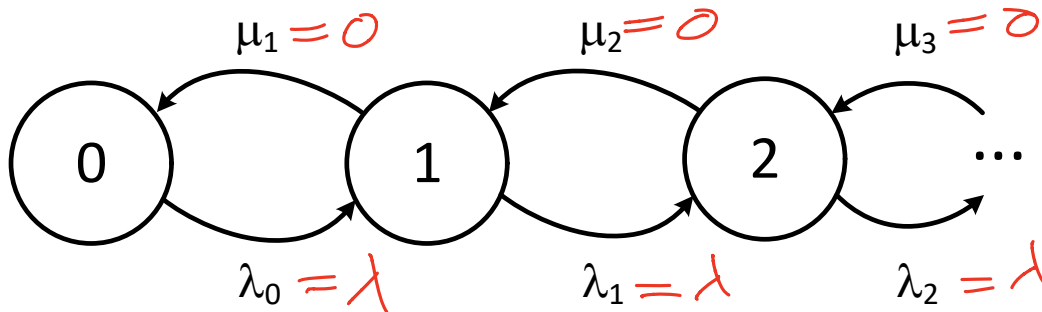
$$\tau, T_B, T_D$$
$$\tau = \min(T_B, T_D)$$



Birth-death processes

- A birth-death process is a CTMC with states $\{0, 1, \dots\}$ for which from state n it is only possible to transition to states $n - 1$ or $n + 1$.
- Consider a birth-death process for which $\mu_n = 0$ and $\lambda_n = \lambda$, for all $n \geq 0$. In this process there are no departures and the inter-arrival times are exponentially distributed with mean $1/\lambda$. What type of process is this one?

Poisson

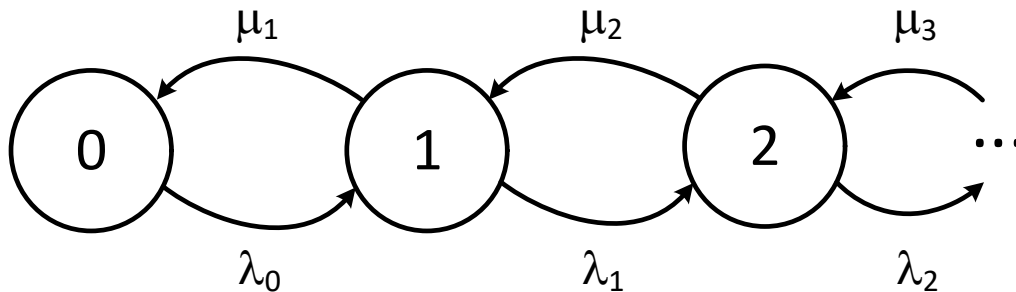


Birth-death processes

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

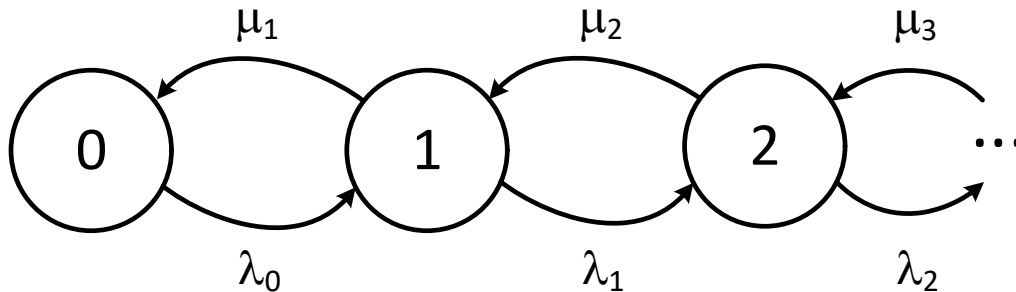
- $v_0 = \lambda_0$
- $v_i = \lambda_i + \mu_i, i > 0$
- $P_{01} = 1$
- $P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i}, i > 0$
- $P_{i,i-1} = \frac{\mu_i}{\lambda_i + \mu_i}, i > 0$

Birth before death
Death " birth



Balance equations of birth-death processes

- State 0: $\lambda_0\pi_0 = \mu_1\pi_1$
- State 1: $(\lambda_1 + \mu_1)\pi_1 = \mu_2\pi_2 + \lambda_0\pi_0$
- State 2: $(\lambda_2 + \mu_2)\pi_2 = \mu_3\pi_3 + \lambda_1\pi_1$
- State $n, n \geq 1$: $(\lambda_n + \mu_n)\pi_n = \mu_{n+1}\pi_{n+1} + \lambda_{n-1}\pi_{n-1}$



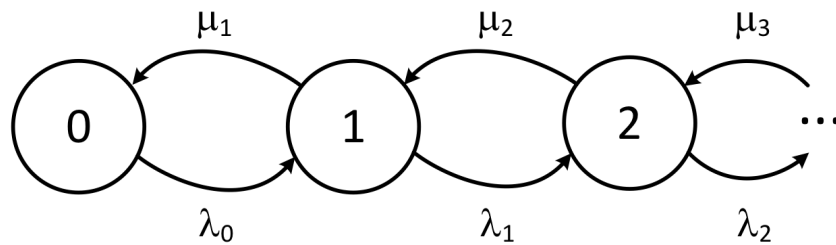
Limiting state probabilities of birth-death processes

$$\pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i}}$$

$$\pi_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \pi_0 = \frac{\frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n}}{1 + \sum_{i=1}^{\infty} \frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i}}, n \geq 1$$

- Necessary and sufficient conditions for the existence of limiting state probabilities:

$$\sum_{i=1}^{\infty} \frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i} < \infty$$



$$0: \lambda_0 \pi_0 = \mu_1 \pi_1$$

$$1: (\lambda_1 + \mu_1) \pi_1 = \mu_2 \pi_2 + \lambda_0 \pi_0$$

$$2: (\lambda_2 + \mu_2) \pi_2 = \mu_3 \pi_3 + \lambda_1 \pi_1$$

⋮

$$n, n \geq 1: (\lambda_n + \mu_n) \pi_n = \mu_{n+1} \pi_{n+1} + \lambda_{n-1} \pi_{n-1}$$

$$* \lambda_0 \pi_0 = \mu_1 \pi_1$$

$$\cancel{\lambda_0 \pi_0} + \cancel{\lambda_1 \pi_1} + \cancel{\mu_1 \pi_1} = \cancel{\mu_1 \pi_1} + \mu_2 \pi_2 + \cancel{\lambda_0 \pi_0}$$

$$* \lambda_1 \pi_1 = \mu_2 \pi_2$$

$$\cancel{\lambda_1 \pi_1} + \cancel{\mu_1 \pi_1} + \cancel{\lambda_2 \pi_2} + \cancel{\mu_2 \pi_2} = \mu_2 \pi_2 + \cancel{\lambda_0 \pi_0} + \mu_3 \pi_3 + \cancel{\lambda_1 \pi_1} = \mu_1 \pi_1$$

$$* \lambda_2 \pi_2 = \mu_3 \pi_3$$

⋮

$$* \lambda_n \pi_n = \mu_{n+1} \pi_{n+1}, \quad n \geq 0$$

$$\lambda_n \pi_n = \mu_{n+1} \pi_{n+1}, \quad n \geq 0$$

$$\pi_1 = \left(\frac{\lambda_0}{\mu_1} \right) \pi_0$$

$$\pi_2 = \left(\frac{\lambda_1}{\mu_2} \right) \pi_1 = \left(\frac{\lambda_1 \lambda_0}{\mu_1 \mu_2} \right) \pi_0$$

⋮

$$\pi_n = \frac{\lambda_{n-1} \lambda_{n-2} \dots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \dots \mu_2 \mu_1} \cdot \pi_0$$

$$\sum_{n=0}^{\infty} \pi_n = 1 \quad \pi_0 + \sum_{n=1}^{\infty} \pi_n = 1$$

$$\pi_0 + \pi_0 \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1} = 1$$

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1}}$$

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1}}$$

$$\pi_n = \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1} \pi_0$$

$$\pi_n = \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1} \cdot \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \dots \lambda_0}{\mu_n \dots \mu_1}}, \quad n \geq 1$$

must converge!